DESIGN VERIFICATION PROTOCOL

Protocol Number: UI01	Title: Usability Verification
Project Number: 1520	Project Name: Intraoral Imager
Usability Specification Number: 4044	PPS Number: SP1-4000

1.0 PURPOSE

To verify that the design of the Digital Impression System conforms with the user interface requirements for primary operating functions listed in SP1-4044, *Digital Impression System Usability Specification*.

2.0 DEVICE DESCRIPTION

The Digital Impression System for Orthodontic Use is a handheld intraoral 3D imager intended for use inside the human oral cavity to digitally capture the three-dimensional topography of teeth, gingiva, and/or palate in clinical settings. It is intended to replace traditional impressions, which are the current standard for recording dental and orthodontic features of patients undergoing orthodontic treatment.

The system is intended to be used by orthodontists and orthodontic assistants in an orthodontic office environment. The system consists of a handheld wand connected to a computer which is housed in a base unit. The computer contains proprietary software to acquire, process, and store the digital 3D image data. Patient information is entered into the software using a touchscreen monitor connected to the computer. To capture a 3D image of the patient's dental arch and/or bite, the operator moves the wand over the surface of the teeth to be scanned. A video camera inside the wand captures images of the teeth surfaces. Algorithms in the software process these images into a 3D image and display the 3D image on the computer monitor in real time. The software also saves the 3D image data and identifying patient information to be used by orthodontic manufacturers to design and manufacture customized orthodontic appliances, (e.g., braces, retainers or aligners).

Indications for Use

The Digital Impression System for Orthodontic Use is an optical impression system intended for use by dental professionals to record the topographical characteristics of teeth, gingiva, and/or palate. The Digital Impression System is intended for use in conjunction with the production of orthodontic appliances, including aligners.

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3.0 REFERENCES

3.1 Standards: EN 62366:2008

3.2 Specifications:

3.2.1 SP1-4044 *Digital Impression System Usability Specification*

3.2.2 SP1-4000 *Product Performance Specification*

4.0 PERSONNEL

4.1 The design will be verified by Product Management, Design Engineers, QA Personnel, or other personnel with expert knowledge of the system design.

5.0 PROCEDURE

5.1 Confirm, through objective evidence, that the system design meets all user interface requirements for the Primary Operating Functions listed in Section 3.3 of SP1-4044 *Digital Impression System Usability Specification*. Document the results of this verification on the attachment. Post the results in the Usability Engineering Folder.

5.2 Record the outcome of each test under Conformance to Specification. Check the Yes box if the design meets the Usability Specification. Check the No box if the design does not meet the usability specification.

5.3 Sign and date each record.

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6.0 REVISION HISTORY

Revision	Date	Description Of Change	Author
1.0	6-26-12	Draft version	D. Hoyle

7.0 APPROVALS

Originator	DPI Engineering	DPI Research & Development	DPI Quality Engineering	DPI Program Management

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8.0 ATTACHMENT

Verification Test Record – Design Conformance to Usability Specification Requirements

By signing below, I confirm that the design meets the requirements of the applicable Usability Specification Number.

Usability Specification Number	Description	Tester Name & Title (Print)	Tester Signature and date	Conformance to Specification	
				Yes	No
UI01	Easy to clean and disinfect system components between each patient.				
UI02	Easy to replace and securely attach the tip support and the disposable tip.				
UI03	One-button power-on/power-off.				
UI04	System check is performed when the system is powered-on (before the first scan).				
UI05	System is lightweight enough to move easily.				
UI06	Touchscreen position is easily adjustable.				

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UI07	Easy to select commands on the touchscreen.				
UI08	Easy to remove the wand from the wand holder.				
UI09	Easy to safely seat the wand in the wand holder.				
UI10	Wand tip fits comfortably in the patient's mouth.				
UI11	Wand is lightweight to enable the operator to move it easily while scanning				
UI12	Easy to create a new case and enter patient information.				
UI13	Operator can successfully scan the patient's lower arch.				
UI14	Operator can successfully scan the patient's upper arch.				
UI15	Operator can successfully run left and right bite scans.				
UI16	Operator can successfully run a Fill scan.				

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UI17	Operator can successfully cancel and repeat a scan segment.				
UI18	Operator can successfully skip a scan segment.				
UI19	Operator can use the touchscreen to inspect images.				
UI20	Operator can successfully complete and approve a digital image.				
UI21	Operator can add, edit, and remove users (must be logged in as an Admin).				
UI22	Operator can add or edit practice information (must be logged in as an Admin).				
UI23	Operator understands that he/she should not stare at the laser beam emitted by the wand, or directly view the beam with an optical device.				